

GISELLE WU

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EDUCATION

Dartmouth College, Hanover, NH

June 2027

B.A. Computer Science and Engineering Sciences | B.E. Computer Engineering

GPA 3.85/4.0

Relevant Coursework: Database Systems, Machine Learning and Statistical Data Analysis, Full-Stack Development, Digital Electronics, Control Theory, Software Design and Implementation, Problem Solving via Object-Oriented Programming, Differential Equations

AIT Budapest, Budapest, Hungary

September - December 2025

Exchange Term, Bachelor of Computer Engineering

Relevant Coursework: Algorithms and Data Structures, Mobile Software Development, Theory of Computing, Computer Graphics

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Kotlin, JavaScript, SQL, MATLAB, R, CSS/HTML

ML/AI: PyTorch, TensorFlow, Transformers, LoRA, RAG, NLP, scikit-learn, NumPy

Systems / DevOps: Linux, AWS, Docker, Git, REST APIs, CI/CD, NVIDIA CUDA

Development: React, Node.js, Express.js, Zustand, Nest.js

Data: PostgreSQL, MySQL, MongoDB, Firebase, Pandas, pgvector

Hardware: FPGA (Basys3), VHDL/Verilog, Arduino, CAD (Solidworks)

Audio/Speech: Deepgram (nova-3-medical STT, aura-2 TTS)

RESEARCH & WORK EXPERIENCE

DALI Lab, Hanover, NH

March 2026 - Present

Fullstack Developer

- Built full-stack AI chatbot with Claude API, injecting live EHR data (labs, vitals, diagnoses) into context for personalized patient Q&A with SSE streaming
- Engineered hybrid rare disease search over 213 entries combining pgvector similarity and lexical scoring with optional HPO/ICD-10CM ontology enrichment
- Integrated Deepgram nova-3-medical STT and TTS for voice accessibility, allowing hands-free clinical interaction

Siemens EDA, Santa Clara, CA

January 2026 - April 2026

Engineering Intern

- Developed LLM-powered log analysis feature in Siemens Aprisa using RAG pipelines, replacing manual triage of large-scale EDA logs with automated structured insights.
- Built a self-healing tool classification pipeline within Siemens Fuse API, reducing irrelevant tool calls by 20% and improving query routing precision across 500+ tools
- Evaluated **NVIDIA CUDA Toolkit** for GPU acceleration by benchmarking fuzzy name-matching workloads, achieving **3x runtime improvement**.

SELECTED PROJECTS

Darthbid

May 2026 - June 2026

- Designed 13-entity relational schema (EERD → MySQL) modeling a course-spot auction marketplace with enrollment transfers, fund settlement, and automatic bid refunds
- Built full-stack auction platform in Flask + React with JWT auth, real-time bid/ask ticker feed, and Recharts price history charts per course section
- Implemented complete transaction lifecycle — list, bid, accept, outbid refund, enrollment transfer, and account ledger — across coordinated backend endpoints

Pet Rock VGA Game

August 2025

- Built Tamagotchi-style FPGA game in VHDL with modular FSMs and real-time VGA graphics
- Debugged **2,000+** LOC and optimized memory (**RGB 4→3 bits**); awarded Top Innovation among 20+ projects

Harmonize

May - June 2025

- Built full-stack web app with **React, Express.js, Leaflet, Zustand, and REST APIs**, integrating geolocation, OpenCage Geocoding API, and Ticketmaster API to display nearby friends and events
- Implemented "Blend Mode" music compatibility algorithm (shared artists/genres) with debounced async calls to minimize redundant API requests during live updates

Tiny Search Engine

November 2024

- Built NLP engine with HMM-based POS tagging, applied Viterbi algorithm to sequence classification
- Trained and tested on Brown Corpus and performed error analysis, identifying impact of domain-specific training bias and unseen-word penalties on model accuracy and generalization

LEADERSHIP EXPERIENCE

Dartmouth College, Hanover, NH

March 2026 - Present

Teaching Assistant, CS 10: Problem Solving via Object-Oriented Programming

- Led weekly recitations for 10+ students in Dartmouth's intro OOP course, covering inheritance, polymorphism, recursion, and design patterns in Java
- Held 5 hours of weekly office hours for 1:1 debugging and code review